

Econometrics Final Project Proposal

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ECON 4050

Purpose

The proposal is due well before the final project so that you can choose a feasible question, identify usable data, and receive feedback before writing the final paper. The main goal is not to have the project finished. The main goal is to show that your question, data, and empirical approach are realistic.

Important data note. For the proposal, you must identify at least three potential datasets. This does not mean your final project must use exactly three datasets. For the final project, you may use one dataset, or as many as you like, depending on what your question requires. Data availability is usually the main constraint, so the proposal asks for multiple possible sources to help you avoid discovering too late that the data cannot support your question.

Observation rule. As a general rule, your analysis dataset should contain at least 1,000 observations. Projects using substantially smaller datasets require instructor approval.

Choose One of Two Proposal Paths

You may submit either a correlational proposal or a causal proposal. Both are acceptable. The most important thing is that the proposal accurately matches the question you can answer with the data and method you plan to use. **Note that you can iterate/ update/ change your idea as you move through the course, but this is intended for you to get the ball rolling so you don't fall behind later.**

Dimension	Path A: Correlational	Path B: Causal
Main question	What is the relationship between X and Y?	What is the effect of X on Y?
Typical goal	Describe an empirical pattern and interpret associations carefully.	Estimate a causal effect using a credible source of variation.
Examples	Marijuana use and violent crime; unemployment and opioid mortality; school spending and test scores.	Marijuana legalization and violent crime; minimum wage changes and employment; Medicaid expansion and health insurance coverage.
Methods likely to fit	OLS, multiple regression, fixed effects, subgroup analysis, descriptive figures.	Difference-in-differences, event studies, regression discontinuity, or instrumental variables.
Key limitation to discuss	Correlation is not causation. Explain omitted variables, reverse causality, and/or selection.	Explain the identifying assumption and why it may fail in your setting.

What to Submit

Submit a PDF proposal that uses either Option A or Option B below. A good proposal should be specific enough that someone could understand your question, find your proposed data, and see how the final project might be completed. This will likely be 2-3 pages.

- Use complete sentences, not just a list of links.

- Include links to each proposed dataset and confirm that the files are downloadable.
- Do not rely on PDFs as your main data source. The data should be available in a format that can be imported into R.
- It is fine if your plan changes after the proposal. The purpose is to start with a feasible plan.

Resources for Finding Data

Useful starting points include government websites, research repositories, and replication archives. When evaluating a dataset, check the unit of observation, time period, geographic coverage, variable definitions, and file format before committing to the topic.

Source type	Examples	Why it may help
Government statistical agencies	Census Bureau, Bureau of Labor Statistics, Bureau of Economic Analysis, CDC, FBI Crime Data Explorer, Department of Education, state or local agencies.	Good for official, regularly updated data with geographic or time-series coverage.
Harvard Dataverse	Search by topic, keyword, paper title, author, or journal.	Good for replication data from published research and cleaned datasets.
ICPSR	Search for surveys, administrative data, criminal justice data, health data, and social science datasets.	Good for well-documented datasets used by social scientists.
AEA data resources and journal replication packages	Use the AEA data page and replication packages from empirical economics papers.	Good for finding data linked to economic research questions.
Other public repositories	Kaggle, university repositories, GitHub repositories, and local open-data portals.	Potentially useful, but verify documentation and data quality carefully.

Proposal Option A: Correlational Path

Use this path if your project will study an empirical relationship without claiming that one variable causes the other.

1. Research Question

State one clear question. For the correlational path, phrase it as a relationship. Example: What is the relationship between marijuana consumption and violent crime across U.S. states?

2. Motivation

Explain why the relationship matters. Identify the economic, policy, or social reason the question is worth studying. Keep this focused and avoid making causal claims at this stage.

3. Expected Data Sources

List at least three potential datasets that can be downloaded in a usable format, such as CSV, Excel, Stata, R, or fixed-width text. For each dataset, include the source, link, unit of observation, years covered, main variables, and whether it can be merged with the other sources.

4. Feasibility Check

Confirm that the datasets are downloadable. Explain any uncertainty you still have.

Suggested Example

- Research question: What is the relationship between marijuana consumption and violent crime across U.S. states?
- Key independent variable: marijuana consumption or prevalence of marijuana use.
- Outcome: state-level violent crime rate.
- Possible approach: regress violent crime rates on marijuana-use measures and controls such as unemployment, income, population, and urbanization.
- Main interpretation issue: the coefficient is an association and may reflect other factors correlated with both marijuana use and crime.

Proposal Option B: Causal Path

Use this path if your project will ask whether a policy, shock, rule, or intervention caused a change in an outcome.

1. Research Question

State one clear causal question. Example: What is the effect of recreational marijuana legalization on violent crime in U.S. states?

2. Motivation

Explain why the causal effect matters. Identify who would care about the answer, why the policy or shock is important, and what prior uncertainty motivates the study.

3. Expected Data Sources

List at least three potential datasets that can be downloaded in a usable format, such as CSV, Excel, Stata, R, or fixed-width text. For each dataset, include the source, link, unit of observation, years covered, treatment or policy variable, outcome variable, control variables, and whether the data can be merged.

4. Feasibility Check

Confirm that the datasets contain enough treated and comparison observations, enough pre- and post-period data if relevant, and enough variation in the treatment. Explain any uncertainty you still have.

Suggested Example

- Research question: What is the effect of recreational marijuana legalization on violent crime in U.S. states?
- Treatment: adoption of recreational marijuana legalization.
- Outcome: state-level violent crime rate.
- Possible design: compare changes in crime before and after legalization in adopting states relative to non-adopting states, while controlling for state and year fixed effects.
- Main identifying concern: states that legalize may differ from non-legalizing states in ways that also affect crime trends.

Final Checklist Before Submitting

- I chose either the correlational path or the causal path.
- My research question is stated clearly and matches the path I chose.
- I identified at least three potential datasets and included links to downloadable files.
- I described the unit of observation, time period, key variables, and file format for each dataset.
- I explained how the datasets could be used together or why one dataset may be sufficient for the final project.
- My analysis dataset should contain at least 1,000 observations. Projects using substantially smaller datasets require instructor approval.